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China: Central Bank Swap to Hong Kong Monetary Authority, 2009¹

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Abstract

Starting in 2008, the People's Bank of China (PBOC) began to grow a global network of central bank swap lines, announcing six in 2008–09 alone. The largest of those lines, in January 2009, was a 200 billion renminbi (RMB; USD 29.2 billion) swap line to the Hong Kong Monetary Authority (HKMA). The PBOC and HKMA, in similarly worded announcements, mentioned two goals of the swap arrangement: to promote renminbi-denominated trade settlement in Hong Kong and to promote financial stability in Hong Kong and the region. Hong Kong was the most important offshore financial center for trading renminbi-denominated securities, and the swap lines enabled the HKMA to promise market participants that it would be able to make renminbi liquidity available in a crisis. In October 2010, in response to tightening offshore renminbi liquidity conditions as the local Hong Kong trade settlement bank reached a PBOC-set conversion quota, the HKMA drew on the line for RMB 20 billion to support renminbi trade settlement in Hong Kong. However, the HKMA ultimately extended no renminbi loans to Hong Kong banks during that episode. The HKMA and PBOC increased the swap line size and extended its terms over time. In 2022, they converted it into the PBOC's first standing swap facility, with an authorized amount of RMB 800 billion.

Keywords: central bank swap line, China, Global Financial Crisis, Hong Kong, renminbi

¹ This case study is part of the Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering central bank swap line programs. Cases are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/>.

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Overview

On December 19, 2008, the central government of the People's Republic of China announced 14 fiscal and monetary measures to support Hong Kong's economy and financial stability during the Global Financial Crisis (GFC). Those measures included a swap line between the People's Bank of China (PBOC) and the Hong Kong Monetary Authority (HKMA). The PBOC and HKMA signed a bilateral currency swap agreement on January 20, 2009. Under the swap arrangement, the HKMA could borrow up to 200 billion renminbi (RMB, also known as yuan; USD 29.2 billion)³ in exchange for Hong Kong dollars (HKD) in a bilateral unidirectional⁴ swap with a three-year term. The PBOC aimed to promote renminbi-denominated trade settlement in Hong Kong and promote financial stability in Hong Kong and the region (HKSAR 2009; HKMA 2009; PBOC 2009a; Wang 2009).

Other central government support efforts included the establishment of a multi-currency payments arrangement with Hong Kong (PBOC 2009b).

Hong Kong's swap line was one of six—and the largest—that the PBOC made available to central banks in 2008 and 2009. The others were with South Korea, Malaysia, Indonesia, Argentina, and Belarus (Bank Negara Malaysia 2009; Dow Jones 2009; Webber 2009; Xinhua News 2009; n.d.).

Key Terms

Purpose: To “bolster investor confidence in Hong Kong's financial stability, promote regional financial stability and the development of yuan-denominated trade settlement between Hong Kong and the mainland” (PBOC 2009a)

Participating Parties	PBOC, HKMA
Type of Swap	Bilateral unidirectional
Currencies Involved	RMB, HKD
Launch Date	January 20, 2009
End Date	Still in effect; became a standing facility on July 4, 2022
Date of First Usage	The HKMA drew on the line in October 2010 for RMB 20 billion
Interest Rate and Fees	Not available
Amount Authorized	RMB 200 billion
Peak Usage Amount and Date	The HKMA had RMB 20 billion outstanding at the end of 2010; it reported increasing amounts in subsequent years
Downstream Use/Application of Swap Funds	No banks borrowed from the HKMA in 2010 when the HKMA first activated the swap line

(continued)

³ According to FRED, USD 1 = RMB 6.84 in January 2009.

⁴ We were unable to access the swap agreement and thus were unable to verify whether the PBOC could draw on the swap line. In at least one other case (Argentina), a PBOC swap agreement was explicitly unidirectional, but in another (Mongolia), it was reciprocal (Arnold 2023a; Arnold 2023b).

On October 28, 2010, as a result of higher-than-expected demand for cross-border trade settlement, the Bank of China Hong Kong—at the time the only official trade settlement clearing bank⁵—reached the PBOC’s RMB 8 billion quota for renminbi conversion, which was a central part of China’s capital control regime (see Appendix A). As a result, offshore renminbi liquidity

Outcomes	The parties to the swap line renewed it five times, and in 2022 it became a standing facility without an express end date
Notable Features	The swap line was one of six the PBOC established during the Global Financial Crisis

dried up, and the onshore-offshore spread⁶ began to widen. In response, the HKMA preemptively activated its PBOC swap line for RMB 20 billion (which represented 10% of the swap cap at the time) and used those funds to create an RMB 20 billion standby credit facility to promote market confidence (HKMA 2011a). However, the HKMA ultimately extended no renminbi loans to Hong Kong banks during that episode (Maziad and Kang 2012). In its 2010 annual report, the HKMA said that as of December 31, 2010, the outstanding swap liability to the PBOC was RMB 20 billion (HKMA 2011b). Based on changes in year-end balances, the HKMA drew on the swap line at least three more times, in 2013, 2014, and 2015, with possible draws occurring in 2016 and 2017⁷ (see Figure 1).

In 2012, the HKMA announced the creation of the Renminbi Liquidity Facility to Renminbi Business Participating Authorized Institutions (RMB Liquidity Facility, or the Facility), which it said would “make use of the currency swap arrangement between the [PBOC] and the HKMA” (HKMA 2012b). The RMB Liquidity Facility provided term renminbi funds to participating authorized institutions against eligible collateral. In its press release database, the HKMA included PBOC swap line press releases under the category “RMB Liquidity Facility” among press releases about the RMB Liquidity Facility (HKMA n.d.b).

In November 2014, the HKMA announced a new Intraday Repo under the RMB Liquidity Facility, through which the HKMA began to offer intraday renminbi funding of up to RMB 10 billion to authorized institutions in Hong Kong⁸ (see Key Design Decision No. 10, Downstream Use of Borrowed Funds) (HKMA 2015a; HKMA 2015b). The HKMA said the Intraday Repo would help Hong Kong banks to manage renminbi payment flow volumes,

⁵ While we were unable to determine its activities in 2010, as of September 2014, the Bank of China Hong Kong was providing intraday renminbi overdraft services to banks participating in renminbi-clearing business in Hong Kong (Chan 2014).

⁶ The renminbi can be traded in mainland China (quoted as CNY) and offshore in overseas exchange centers (historically Hong Kong and Singapore) (quoted as CNH). Since the onshore renminbi exchange rate is more tightly controlled by the PBOC, the onshore-offshore spread is sensitive to demand fluctuations in offshore markets (Maziad and Kang 2012).

⁷ Stock amounts do not necessarily represent draw amounts. For example, the HKMA could draw RMB 50 billion in January and pay RMB 20 billion in July, resulting in a year-end stock amount of RMB 30 billion. Further, stock amount does not correspond to number of draws; for example, the HKMA could make one draw of RMB 50 billion or five draws of RMB 10 billion, resulting in the same year-end stock amount. We were unable to verify whether small changes (in 2016 and 2017) were the result of interest accumulation or small draws on the swap line.

⁸ In the same year, renminbi trade settlement in Hong Kong increased by 63%, and renminbi bond issuance increased by 69% (HKMA 2015b).

which would in turn strengthen the infrastructure of Hong Kong's offshore renminbi market. On the same day it announced the Intraday Repo, the HKMA announced the designation of seven CNH⁹ primary liquidity providers (PLPs), which pledged to expand their market-making roles in offshore renminbi products markets in exchange for a dedicated RMB 2 billion repo facility (per institution) (HKMA 2014b). That same year (2014), the year-end outstanding obligations under the swap largely ceased to be round numbers (see Figure 1). Between 2016 and 2022, Hong Kong banks used the RMB Facility (including intraday and overnight repo facilities as well as the dedicated PLP facility) for trillions of renminbi borrowing, with aggregate usage of the Facility generally increasing over time (HKMA 2023).

The two central banks renewed the swap line in 2011, 2014, 2017, 2020, and 2022. In 2022, they upgraded it to a standing facility with an RMB 800 billion cap (HKMA 2011a; HKMA 2014c; HKMA 2017a; HKMA 2020b; HKMA 2022a). Figure 1 shows the size and term changes of the swap line.

Figure 1: PBOC-HKMA Bilateral Swap Line, 2009–2022

Year	Swap Cap (RMB billions)	Year-end Swaps Outstanding (RMB billions)	YoY Change in Swap Outstanding (RMB billions)	Term (years)
2009	200	0	Not Applicable	3
2010	200	20.0	20.0	3
2011*	400	20.0	0	3
2012	400	0	(20.0)	3
2013	400	40.0	40.0	3
2014*	400	51.1	11.1	3
2015	400	55.6	4.5	3
2016	400	56.0	0.4	3
2017*	400	56.8	0.8	3
2018	400	50.0	(6.8)	3
2019	400	0	(50.0)	3
2020*	500	0	0	5
2021	500	0	0	5
2022*	800	Not Available	Not Available	Standing Facility

Note: Years in which the swap line was renewed are indicated with an asterisk.

Sources: HKMA 2011b; HKMA 2013; HKMA 2015b; HKMA 2017b; HKMA 2019; HKMA 2021; HKMA 2022c.

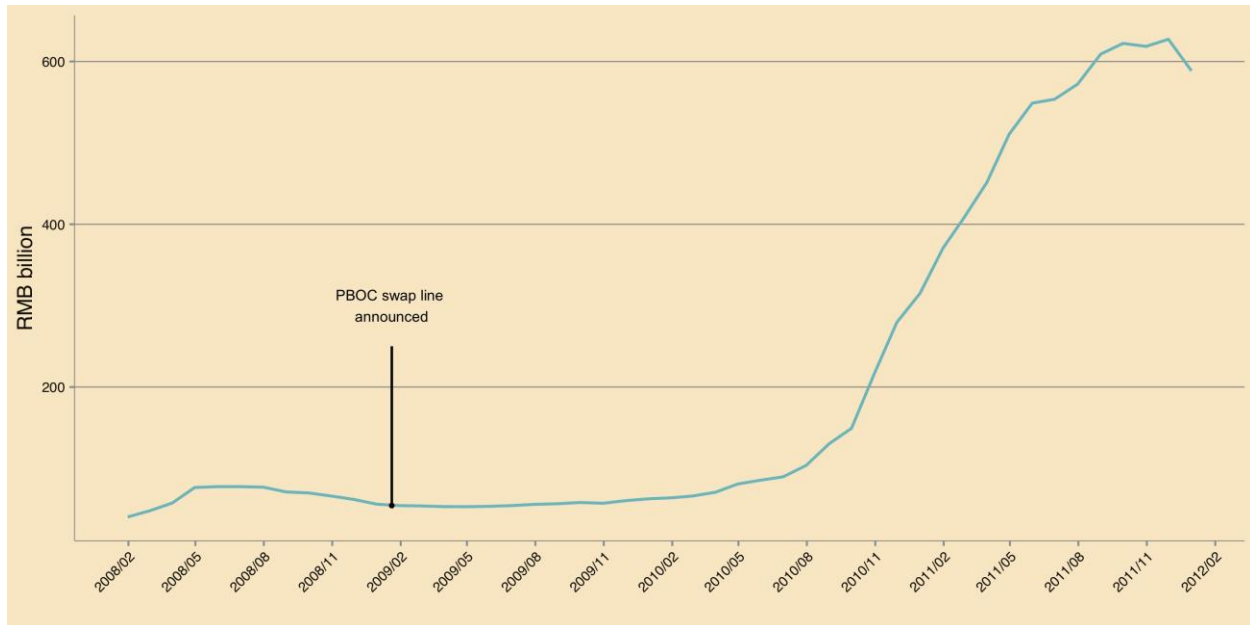
Summary Evaluation

The official aims of the swap line were to promote renminbi-denominated trade settlement in Hong Kong and promote financial stability (HKSAR 2009; HKMA 2009; PBOC 2009a; Perks et al. 2021; Wang 2009). The chief executive of the HKMA, Norman Chan, noted in 2011 upon the extension of the swap line that “the development of renminbi business in Hong Kong has made very encouraging progress over the past year” (HKMA 2011a).

⁹ CNH refers to the offshore renminbi.

Hong Kong was the most important offshore financial center for trading renminbi-denominated securities, and the swap lines enabled the HKMA to promise market participants that it would be able to make renminbi liquidity available in a crisis (McDowell 2019). In 2010–11, renminbi deposits and banking activity increased in Hong Kong. Figure 2 shows total renminbi deposits in Hong Kong banks between 2008 and year-end 2011. Starting in early 2010, total deposits began a sharp increase. They reached RMB 622 billion, more than 10% of total deposits in Hong Kong, in September 2011 (Maziad and Kang 2012).

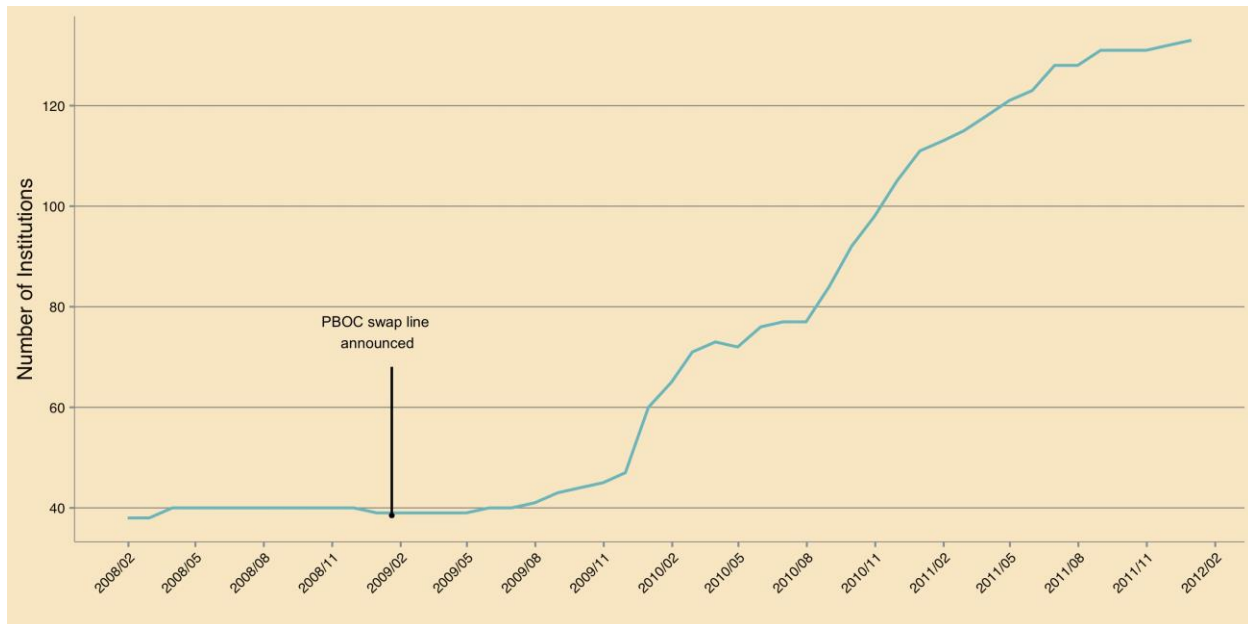
Figure 2: Total Renminbi Deposits in Hong Kong Licensed Banks



Source: Bloomberg.

Figure 3 shows the number of Hong Kong financial institutions engaged in renminbi transactions.

Figure 3: Number of Authorized Hong Kong Institutions Engaged in Renminbi Banking Business



Source: Bloomberg.

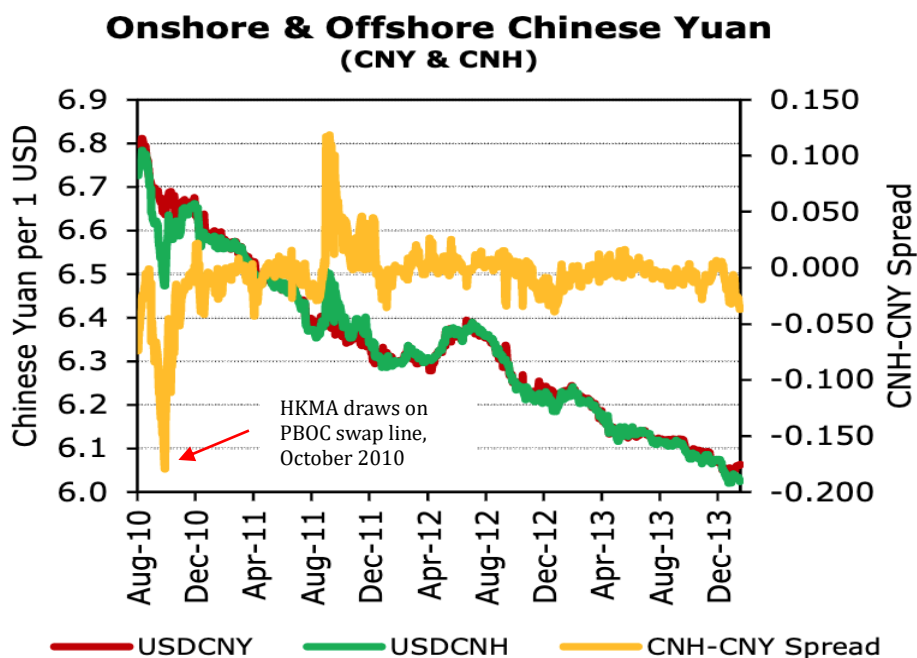
Some scholars have said that the Chinese central government used the swap lines with the HKMA and other central banks to promote the internationalization of the renminbi. They note several motives: to gain political leverage, promote trade, reduce reliance on the US dollar, reduce foreign exchange risk, and limit the accumulation of foreign reserves (Cao 2022; Gao and Yu 2012; Maziad and Kang 2012; McDowell 2019; Perks et al. 2021; Zhang 2015). Hong Kong had long been the central government's preferred point of contact with the international financial system for renminbi internationalization (Leung 2011). One scholar called the PBOC swap lines a form of "financial statecraft" (McDowell 2019). A former Bank of England official has said that "China's objectives are not to do with crisis management, but promoting the use of the [renminbi] in international trade" (China Daily 2014). Other scholarship has shown that trade is a key determinant in Chinese swap line extension and that China's swap lines effectively promote trade with China when deployed (Liao and McDowell 2015; Perks et al. 2021).

A scholar of China's swaps has said that the PBOC's swaps to offshore renminbi markets—such as Hong Kong—serve a distinct purpose compared with its other swap lines in that the swaps ensure renminbi liquidity, thus promoting confidence in, and growing, offshore renminbi markets. In other words, the PBOC's swap lines to the European Union, United Kingdom, Singapore, and Hong Kong serve as liquidity insurance for nascent offshore renminbi banking and bond markets (McDowell 2019). In its 2014 annual report discussing the swap line renewal, the HKMA said that the swap would allow it to provide necessary liquidity so as to "maintain the stability of the offshore renminbi market in Hong Kong" (HKMA 2015b).

Some observers have also noted that the central government had provided comparatively cheap access to the renminbi in Hong Kong, which resulted in a surge in speculative offshore demand for the currency in Hong Kong (Yung 2010; Zhang 2015). The swap line effectively served as a release valve for speculative pressure on the onshore renminbi; the onshore-offshore spread narrowed sharply after the HKMA drew on the swap (Maziad and Kang 2012; Ro and Labuszewski 2014).

Figure 4 shows the onshore-offshore renminbi spread around the time of the 2010 swap line draw.

Figure 4: Onshore & Offshore Renminbi Spread, August 2010–December 2013



Source: Ro and Labuszewski 2014.

Context: Hong Kong SAR, China 2009–2010		
GDP (SAAR, nominal GDP in USD)		\$214.0 billion in 2009
		\$228.6 billion in 2010
GDP per capita (SAAR, nominal GDP in USD)		\$30,697 in 2009
		\$32,550 in 2010
Sovereign credit rating (five-year senior debt)	2009	Moody's: Data not available
		S&P: Data not available
		Fitch: Data not available
	2010	Moody's: Data not available
		S&P: Data not available
		Fitch: Data not available
Size of banking system		\$416.8 billion in 2009
		\$523.3 billion in 2010
Size of banking system as a % of GDP		194.7% in 2009
		228.9% in 2010
Size of banking system as a % of financial system		Data not available in 2009
		Data not available in 2010
Five-bank concentration of banking system		80% in 2009
		80.5% in 2010
Foreign involvement in banking system		92% in 2009
		92% in 2010
Existence of deposit insurance		Yes in 2009
		Yes in 2010
Sources: Bloomberg; World Bank Global Financial Development Database; World Bank Deposit Insurance Dataset.		

Key Design Decisions

- 1. Purpose and Type: The PBOC entered into a bilateral swap line with the HKMA to develop renminbi-denominated trade in Hong Kong and promote financial stability in Hong Kong and the region.**

The PBOC aimed to promote renminbi-denominated trade settlement in Hong Kong and promote financial stability in Hong Kong and the region (Cao 2022; HKSAR 2009; HKMA 2009; PBOC 2009a; Wang 2009). In 2020, the HKMA said that the purpose of the (2009) swap line was to promote renminbi trade in Hong Kong (HKSAR 2020). In October 2010, the HKMA activated the swap line for RMB 20 billion, to support renminbi trade settlement after renminbi demand exceeded the conversion quota of the Bank of China Hong Kong—at the time the only official trade settlement clearing bank. For further information about the quota and the Bank of China Hong Kong’s role in renminbi conversion, see Appendix A. The authorities said that the swap line was not needed, but its use “ensured market confidence” (HKMA 2011a; HKMA 2011b; Maziad and Kang 2012).

- 2. Part of a Package (A): The swap line was one of several central bank and government actions the central Chinese government took during the Global Financial Crisis to support Hong Kong’s economy and financial stability.**

On December 19, 2008, the central government announced 14 monetary and fiscal measures to support Hong Kong’s economy and financial stability, including a swap line with the HKMA. The PBOC and HKMA signed a bilateral currency swap agreement on January 20, 2009. Other measures announced included encouragement of mainland branch office development in Hong Kong and the advancement of numerous infrastructure projects (HKSAR 2009; HKMA 2009; PBOC 2009a).

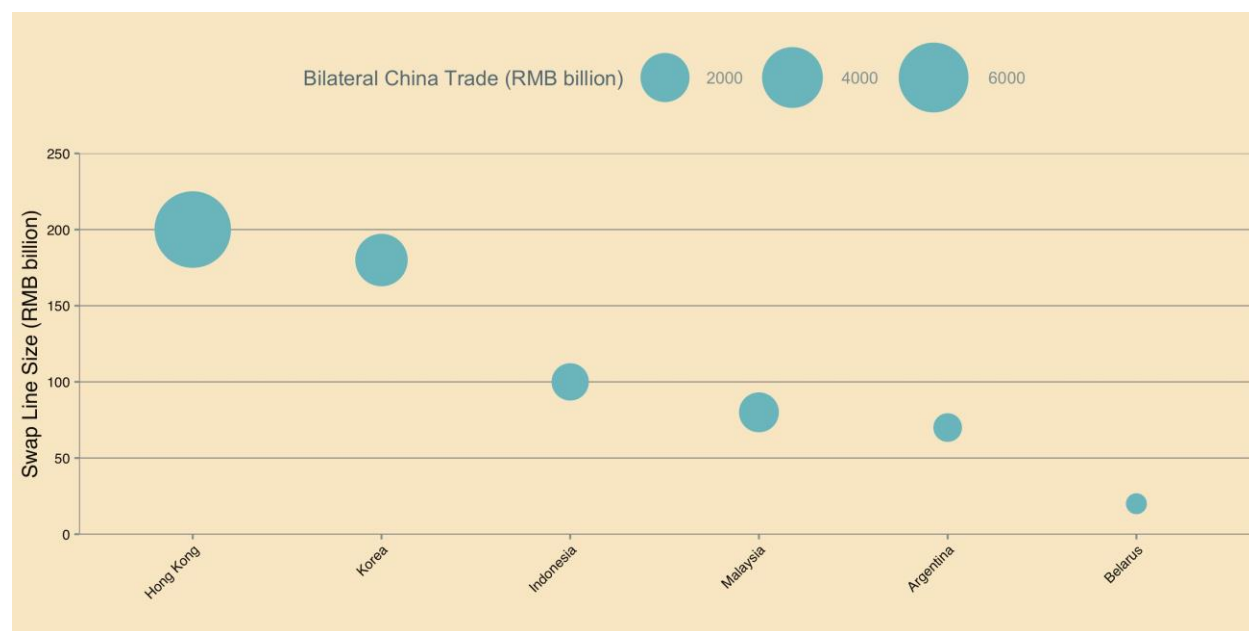
On March 12, 2009, the PBOC announced the establishment of a multicurrency payments arrangement with Hong Kong (PBOC 2009b). On March 16, 2009, the PBOC and HKMA signed a memorandum of understanding with respect to a cross-border payment interoperability arrangement (Wang 2009). On June 29, 2009, the PBOC and HKMA signed a second “supplementary memorandum of co-operation” on the cross-border payment arrangement (Holland 2009). In July, the central government officially announced the renminbi trade settlement pilot scheme. In August, mainland banks issued two floating-rate renminbi bonds in Hong Kong, and in October, the central government Ministry of Finance issued RMB 6 billion in sovereign bonds in Hong Kong (IMF 2009).

Part of a Package (B): The swap line was one of six central bank swap lines extended by the PBOC in 2008–09.

The PBOC-HKMA swap line was part of—and the largest within—a network of six swap lines that the PBOC made available to other central banks during the Global Financial Crisis in 2008–09 (Bank Negara Malaysia 2009; Dow Jones 2009; Webber 2009; Xinhua News n.d.; Xinhua News 2009). Figure 5 shows the sizes of swap lines extended by the PBOC to various

central banks around the world in 2008 and 2009 and compares them to the amount of Chinese exports to those countries. The size of a country's swap line (represented by the y-axis) closely tracked the value of Chinese exports to that country (represented by the size of the bubbles). In other words, the larger the exports from China to the recipient country, the larger the swap line.

Figure 5: Size of Bilateral Swap Lines Extended from the PBOC to Central Banks, 2008–2009



Note: The size of the swap line correlated closely with the value of Chinese exports to the recipient country (represented by the bubbles). In other words, the larger the exports from China to the recipient country, the larger the swap line. (This was a simple correlation using the size of the swap lines and Chinese exports data for the six countries listed in the figure [World Bank 2022a].)

Sources: Bank Negara Malaysia 2009; Dow Jones 2009; Webber 2009; Xinhua News n.d.; Xinhua News 2009.

Part of a Package (C): The HKMA and PBOC were members of other regional networks of international liquidity lines.

The HKMA maintained a network—of which the PBOC was a member¹⁰—of regional US Treasuries-backed bilateral currency repurchase lines since 1996 (HKMA n.d.a; Whai 1996). Under the facility, the PBOC could purchase US Treasury securities from the HKMA—providing the HKMA with renminbi liquidity—and agree to sell them back at a later date; likewise, the HKMA could purchase from the PBOC, reversing the arrangement (HKMA n.d.a; HKMA 2011b).

¹⁰ Other members included Australia, Indonesia, Japan, Malaysia, New Zealand, the Philippines, Singapore, South Korea, and Thailand (HKMA n.d.a).

On March 24, 2010, the Chiang Mai Initiative Multilateralization (CMIM) came into effect, establishing a network—of which the PBOC and HKMA were members—of short-term US dollar swaps (HKMA 2011b). (For a discussion of the CMIM, see Hoffner 2023.)

Figure 6 shows Hong Kong’s international liquidity lines from 2008–2010.

Figure 6: Hong Kong’s International Liquidity Arrangements, 2008–2010

Country	Bilateral Repo Line	Bilateral Swap Line	CMIM
Australia	•		
Brunei			•
Cambodia			•
China	•	•	•
Indonesia	•		•
Japan	•		•
Laos			•
Malaysia	•		•
Myanmar			•
New Zealand	•		
Philippines	•		•
Singapore	•		•
South Korea	•		•
Thailand	•		•
Vietnam			•

Source: HKMA 2011b.

3. Legal Authority: Chinese law gives the PBOC the authority to handle international financial activities.

The PBOC’s enabling law gives it the authority to issue the renminbi, dictate and implement monetary policy, regulate foreign exchange markets, and engage in “relevant international financial activities” (Law of the PRC on the PBOC 2003, art. 4). Any nonmonetary policy decisions must be approved by the State Council, China’s highest administrative governing body¹¹ (Law of the PRC on the PBOC 2003).

Section 3 of the Exchange Fund Ordinance (the Ordinance) authorizes the creation of the Exchange Fund and delegates authority over that fund to the Financial Secretary (LegCo 2019, sec. 3[1]). The Ordinance authorizes the Financial Secretary “with a view to maintaining Hong Kong as an international financial centre, [to] use the Fund as he thinks fit to maintain the stability and integrity of the monetary and financial systems of Hong Kong” (LegCo 2019, sec. 3[1A]). Section 3 of the Ordinance authorizes the Financial Secretary to use the Exchange Fund to enter into foreign exchange transactions and “any financial arrangement that he considers appropriate” (LegCo 2019, sec. 3[2][a]-[b]). Section 5A of the

¹¹ The State Council is the equivalent to China’s executive cabinet; the head of the State Council is the premier, who is commonly the second most powerful member of the ruling Chinese Communist Party. It is primarily in charge of running the “day-to-day affairs of the state” (Wright 2015, 32).

Ordinance stipulates that the Financial Secretary shall appoint a Monetary Authority and section 5B authorizes the Financial Secretary to delegate powers conferred or imposed under the Ordinance to the Monetary Authority (LegCo 2019, sec. 5[A]-[B]).

4. Governance: The State Council approved the 2022 conversion of the swap line into a standing agreement.

The State Council approved the 2022 conversion of the PBOC-HKMA swap line into a standing agreement and the increase in its size from RMB 500 billion/HKD 590 billion to RMB 800 billion/HKD 940 billion (Cao 2022). In the absence of the swap agreement, further governance details were unavailable.

5. Administration: Neither the PBOC nor the HKMA provided any details regarding the low-level management of the swap line.

Neither the PBOC nor the HKMA provided details regarding the low-level management of the swap line.

6. Communication: The PBOC and HKMA, in similarly worded press releases, communicated that the swap line was meant to both provide liquidity support and expand RMB trade; in October 2010, the HKMA announced that it was drawing on the line to provide preemptive renminbi liquidity and used the drawn funds to set up a standby credit facility for Hong Kong banks.

The PBOC and HKMA announced the swap line in similarly worded press releases on January 20, 2009. Both press releases said that short-term liquidity support—both for Chinese banks operating in Hong Kong and Hong Kong banks operating in China—and the expansion of renminbi-based trade were primary goals of the swap line. The press releases did not include details with respect to the administration, governance, fees, or other details; and the text of the swap agreement was not included in the announcement (HKMA 2009; PBOC 2009a).

In its 2010 annual report, the HKMA said that it had activated its PBOC swap line “for facilitating trade settlement business” (HKMA 2011b). In a working paper, two IMF economists said that “although no banks actually needed to tap RMB funding through the swap agreement, its activation as a precautionary measure ensured market confidence” (Maziad and Kang 2012).

In its press release database, the HKMA included PBOC swap line press releases under the category “RMB Liquidity Facility” among press releases about the RMB Liquidity Facility (HKMA n.d.b). The HKMA made public daily usage data for its RMB Liquidity Facility through an application programming interface (a data query tool) (HKMA 2023).

7. Eligible Institutions: The HKMA was the only institution eligible for this bilateral swap line.

The HKMA was the only institution eligible for this specific bilateral swap line, but the PBOC extended a number of central bank swap lines in 2008–09 (see Key Design Decision No. 2B, Part of a Package).

8. Size: The initial line was for RMB 200 billion.

In 2009, the PBOC authorized RMB 200 billion for a term of three years (HKMA 2009; PBOC 2009a). The PBOC and HKMA renewed the swap line in 2011, 2014, 2017, and 2020. In 2022, they upgraded it to a standing facility with an RMB 800 billion cap (HKMA 2011a; HKMA 2014c; HKMA 2017a; HKMA 2020b; HKMA 2022a). (See Figure 1 at Overview.)

9. Process for Utilizing the Swap Agreement: We were unable to uncover details of the implementation process.

We were unable to access the swap agreement and thus were unable to uncover details of the implementation process. Usually, a standard swap agreement involves the borrowing central bank exchanging its own currency (or a “hard” reserve currency) for an equal amount of the lending central bank’s currency; the two banks then swap back the currencies at a future date (McDowell 2019; Perks et al. 2021). In other PBOC swap line cases, the borrowing central bank would request a draw on the swap line and make a purchase transaction of renminbi with local currency, which it would later unwind with a repurchase transaction of its local currency with renminbi (Arnold 2023a; 2023b).

10. Downstream Use of Borrowed Funds: The HKMA drew on the PBOC swap line in 2010 after the Bank of China Hong Kong hit its conversion quota.

In its 2010 annual report, the HKMA said that it had activated its PBOC swap line “for facilitating trade settlement business” (HKMA 2011b). In October 2010, as a result of higher-than-expected demand for cross-border trade settlement, the Bank of China Hong Kong—at the time the only official trade settlement clearing bank—reached the RMB 8 billion quota for renminbi conversion. As a result, offshore renminbi liquidity dried up and the onshore-offshore spread¹² began to widen (Maziad and Kang 2012). In response, the HKMA preemptively activated its PBOC swap line for RMB 20 billion (which represented 10% of the swap cap at the time) to promote market confidence (HKMA 2011a).

In 2012, the HKMA announced the creation of the RMB Liquidity Facility, which it said would “make use of the currency swap arrangement between the [PBOC] and the HKMA” (HKMA 2012b). The Facility was funded from draws on the HKMA’s swap line with the PBOC (Cockerell and Shoory 2012). The RMB Liquidity Facility provided renminbi funding for a

¹² The renminbi can be traded in mainland China (quoted as CNY) and offshore in overseas exchange centers (historically Hong Kong and Singapore) (quoted as CNH). Since the onshore renminbi exchange rate is more tightly controlled by the PBOC, the onshore-offshore spread is sensitive to demand fluctuations in offshore markets (Maziad and Kang 2012).

tenor of one week at an interest rate benchmarked to market rates against eligible collateral (which comprised bills and notes from the Exchange Fund, HKSAR government bonds, and Hong Kong-issued renminbi-denominated Chinese government bonds); the facility imposed haircuts on collateral¹³ (HKMA 2012a).

In November 2014, the HKMA announced a new Intraday Repo under the RMB Liquidity Facility, through which the HKMA began to offer intraday renminbi funding of up to RMB 10 billion to authorized institutions in Hong Kong. Under the Intraday Repo, the HKMA would provide overnight renminbi funding against an expanded list of eligible collateral¹⁴ for a fee benchmarked to market overnight rates¹⁵ (HKMA 2015a; HKMA 2015b). For details of the RMB Liquidity Facility and Intraday Repo as of the November 2014 announcement, see Appendix B.

On the same day it announced the Intraday Repo, the HKMA announced the designation of seven CNH primary liquidity providers (PLPs), which pledged to expand their market-making roles in offshore renminbi products markets in exchange for a dedicated RMB 2 billion repo facility (per institution). The HKMA said the designation scheme was “an important step for further enhancing the infrastructure for the offshore renminbi market in Hong Kong” (HKMA 2014). Among the seven entities designated was the Bank of China Hong Kong,¹⁶ the same bank that in 2010 reached its renminbi-conversion quota, triggering the first draw on the HKMA-PBOC swap line (HKMA 2014b). The HKMA reviewed its list of designated PLPs every two years. Figure 7 shows the PLPs.

¹³ Applicable haircuts were 5% for securities with a maturity of 1 year or less; 10% for securities with a maturity of 1–5 years; and 20% for securities with a maturity of more than 5 years (HKMA 2012a).

¹⁴ The HKMA expanded eligible collateral to include renminbi-denominated bonds issued by Chinese policy banks in Hong Kong (for a list of those banks, see Appendix B) (HKMA 2015a).

¹⁵ In its daily RMB Liquidity Facility usage data, the HKMA presents the Facility as comprising three distinct facilities: (1) the Intraday Repo, (2) the Overnight Repo, and (3) the PLP Facility (HKMA 2023).

¹⁶ As of September 2014, the Bank of China Hong Kong was providing intraday renminbi overdraft services to banks participating in renminbi-clearing business in Hong Kong (Chan 2014). Then chief executive of the HKMA Norman Chan said that the HKMA’s RMB Liquidity Facility would provide intraday repo services “in addition” to the overdraft services being provided by the Bank of China Hong Kong (Chan 2014).

Figure 7: HKMA-Designated CNH Primary Liquidity Providers, as of 2023

Bank	First Designated
Agricultural Bank of China Limited	October 27, 2016
Bank of China (Hong Kong) Limited	October 27, 2016
Bank of Communications Co., Ltd.	November 3, 2014
BNP Paribas	November 3, 2014
China Construction Bank (Asia) Corporation Limited	November 3, 2014
Citibank, N.A.	November 3, 2014
Hongkong and Shanghai Banking Corporation Limited	November 3, 2014
Industrial and Commercial Bank of China (Asia) Limited	November 3, 2014
Standard Chartered Bank (Hong Kong) Limited	November 3, 2014

Note: The HKMA last designated all of the banks on October 26, 2022.

Source: HKMA 2014b; HKMA 2016; HKMA 2018; HKMA 2020a; HKMA 2022b.

The HKMA's RMB Liquidity Facility recorded relatively high usage between 2016 and 2022. The PLP facility, which in 2016 represented RMB 18 billion (RMB 2 billion per each of the nine designated PLPs), recorded average daily use exceeding RMB 18 billion¹⁷ in four of the seven years from 2016 to 2022 (see Figure 8, showing average daily use for repo facilities under the RMB Liquidity Facility). In every year, the Overnight Repo facility was used less than the other facilities.

Figure 8: Daily Average Facility Use by Type (RMB billions), 2016–2022

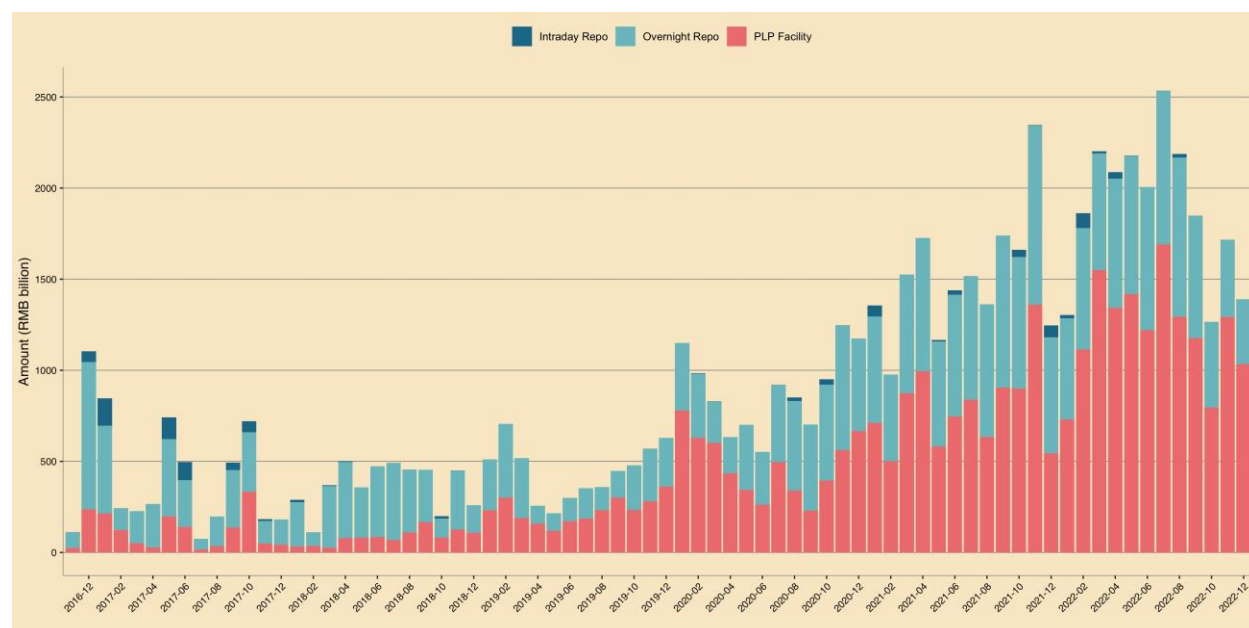
Year	Intraday Repo	Overnight Repo	PLP Facility
2016 ^A	6.2	1.4	21.3
2017	5.5	1.9	11.4
2018	4.0	0.2	13.7
2019	11.2	0.0	10.5
2020	23.1	0.2	19.7
2021	38.7	0.8	33.4
2022	59.6	0.7	31.5

^A Data for 2016 only included daily data from November 1 through the end of the year.

Source: HKMA 2023.

The HKMA's RMB Liquidity Facility experienced usage in the trillions of renminbi from 2016 to 2022, with aggregate usage roughly increasing over time (see Figure 9, showing usage of the RMB Liquidity Facility by facility type over time). The Intraday Repo was consistently used the least of the three, while the PLP's share of total facility usage increased over time.

¹⁷ The HKMA did not disclose how daily limits could have exceeded the facility limit. It is possible that banks could draw RMB 2 billion per draw but could roll draws.

Figure 9: RMB Liquidity Facility Use by Type, 2016–2022

Source: HKMA 2023.

11. Duration of Swap Draws: We were unable to determine the duration of draws under the swap.

Since we were unable to access the swap agreement and information about specific swap draws, we were unable to determine the duration of draws under the swap.

12. Rates and Fees: Neither the PBOC nor the HKMA publicly disclosed information about rates or fees.

Neither the PBOC nor the HKMA publicly disclosed information about rates or fees.

13. Balance Sheet Protection: Since we were unable to access the swap agreement, we were unable to determine any specific balance sheet protection measures.

Since we were unable to access the swap agreement, we were unable to determine any specific balance sheet protection measures. The PBOC in 2009 maintained foreign currency reserves worth more than five times total Chinese external debt. In 2008, the PBOC pegged the renminbi to the US dollar, after which it remained stable against the US dollar in 2009 (Treasury 2009; World Bank 2022b). A standard swap agreement involves the borrowing central bank exchanging its own currency for an equal amount of the lending central bank's currency; the two banks then swap back the currencies at a future date—in essence, the borrowing central bank's currency acts as collateral against the loaned currency (Perks et al.

2021). In other PBOC swap lines, the PBOC held the counterparty central bank's currency as collateral.¹⁸

14. Other Restrictions: Since we were unable to access the swap agreement, we were unable to determine which, if any, restrictions applied to the PBOC's swap line to the HKMA.

Since we were unable to access the swap agreement, we were unable to determine which, if any, restrictions applied to the PBOC's swap line to the HKMA.

15. Other Options: We were unable to uncover other options considered by the PBOC to provide renminbi liquidity to the HKMA.

We were unable to uncover other options considered by the PBOC to provide renminbi liquidity to the HKMA. However, the PBOC and HKMA maintained two other international liquidity lines from 2008 to 2010 (see Key Design Decision No. 2C, Part of a Package).

16. Exit Strategy: The PBOC and HKMA renewed the swap line and eventually converted it into an RMB 800 billion standing facility.

The swap line had an original three-year term but was renewed by both banks in 2011 (HKMA 2009; HKMA 2011a; PBOC 2009a). After multiple additional renewals, the line was converted to a standing facility in 2022 with an RMB 800 billion cap (State Council 2022). (See Figure 1 at Overview.)

¹⁸ For example, see Arnold 2023a; Arnold 2023b.

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Appendixes

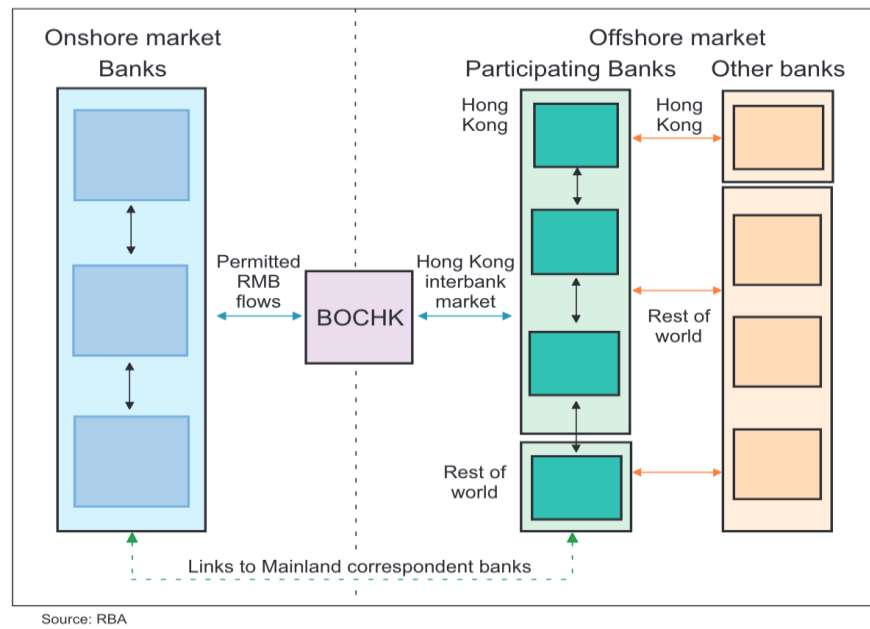
Appendix A: Background of Renminbi Conversion and the PBOC's Capital Controls

Historically, China has employed capital account regulations to restrict the convertibility of the renminbi by limiting fund flows between onshore and offshore markets (Cockerell and Shoory 2012). The Chinese government has employed such capital controls for numerous reasons, including to avoid capital flight, to manage the exchange rate of the renminbi, and to control the allocation of domestic savings¹⁹ (Lee 2021).

Starting in July 2009, the PBOC began to reform renminbi convertibility via the liberalization of offshore renminbi markets through the pilot renminbi trade settlement scheme. Offshore renminbi convertibility liberalization took two forms: (1) loosening of restrictions on the conversion of foreign currencies for renminbi in order to settle trade-related transactions (related to trade, or the current account)—the object of the Pilot RMB Trade Settlement Scheme—and (2) loosening of restrictions on the conversion of foreign currencies related to capital flows (related to investment, or the capital account), which comprised foreign direct investments and portfolio investments (Cockerell and Shoory 2012).

The only banks permitted to serve as the link between the onshore and offshore renminbi markets were the Bank of China Hong Kong (BOCHK) and the Bank of China Macau. Since the BOCHK was the only renminbi-clearing bank in Hong Kong, the PBOC's conversion restrictions for Hong Kong were applied to the BOCHK (Cockerell and Shoory 2012). In other words, the vehicle of the PBOC's offshore-conversion capital controls (in Hong Kong) was a conversion quota on the BOCHK, which controlled all renminbi conversion from Hong Kong (outside of the HKMA in exigent circumstances). Figure 10 shows the BOCHK's role in the offshore renminbi market.

¹⁹ China's state-led investment growth model requires that the state be able to direct investment. It does so through the allocation of domestic savings in state banks, for which there are no viable alternatives in the framework of strict capital controls (Lee 2021).

Figure 10: The Bank of China Hong Kong's Role in the Offshore Renminbi Market

Source: Cockerell and Shoory 2012.

Figure 11 shows the PBOC's renminbi convertibility liberalizations over time.

Figure 11: Renminbi Convertibility Reform, 2003–2014

1. Trade settlement and offshore market	
RMB trade settlement	
Jul 2009	Initial pilot scheme between five Mainland cities and Hong Kong, Macau and ASEAN
Jun 2010	Expanded scheme for trade between 20 Mainland provinces and rest of the world
Mar 2012	Expanded to cover all trade with China
Offshore RMB market	
Feb 2004	Hong Kong banks permitted to offer RMB personal accounts to residents
May 2009	HSBC and Bank of East Asia are first foreign banks to gain approval to issue offshore RMB (dim-sum) bonds
Feb 2010	Foreign firms allowed to issue RMB (dim-sum) bonds
Jul 2010	All corporates allowed to hold RMB accounts and RMB effectively made convertible in offshore market
Aug 2010	McDonald's issues first dim-sum bond by a multinational non-financial corporate
Apr 2011	The first offshore RMB-denominated IPO (by Chinese property investment trust, Hui Xian)
Oct 2011	Baosteel becomes first non-bank Chinese company to issue dim-sum bonds directly
Jan 2012	First approval for a Mainland company to borrow RMB directly from an offshore bank
Jun 2012	HKMA launches RMB liquidity facility to Participating Banks ^(a) in Hong Kong
2. Capital flows	
Inward flows	
Jan 2003	Qualified Foreign Institutional Investor (QFII) scheme for foreign investment in listed Mainland bonds and equities
Aug 2010	Scheme to allow foreign central banks, offshore RMB clearing banks and Participating Banks to invest RMB raised offshore in the Mainland interbank bond market
Oct 2011	Rules formalised to allow approved foreigners to invest RMB raised offshore directly in Mainland firms, including through the provision of RMB cross-border loans
Dec 2011	RMB Qualified Foreign Institutional Investor (RQFII) scheme allowing RMB raised offshore to be invested in listed Mainland bonds and equities
Apr 2012	QFII and RQFII quotas expanded
May 2012	Rules formalised for onshore non-financial corporations to issue offshore RMB bonds
Outward flows	
Apr 2006	Qualified Domestic Institutional Investor (QDII) program launched, allowing domestic institutions to convert RMB into foreign currency and invest in overseas equities and bonds.
Jan 2011	Mainland firms allowed to apply to take RMB offshore for overseas direct investment (ODI) in foreign firms
3. Onshore market	
Apr 2012	PBC widened the daily trading band for the USD/CNY exchange rate to 1 per cent above or below the reference rate
By 2014	Chinese International Payments System (CIPS) to be developed
(a) Participating Banks are those banks with an agreement with the Bank of China (Hong Kong) (BOCHK); they have direct access to the offshore interbank RMB market and are able to undertake cross-border RMB settlement via the BOCHK	
Sources: various official sources, media and market reports	

Source: Cockerell and Shoory 2012.

Appendix B

Figure 12: Terms and Conditions of the 1-Day and 1-Week RMB Liquidity Facility

	Tomorrow/1-Week (1-week funds available next day)	Tomorrow/Next (1-day funds available next day)	Overnight (1-day funds available same day)
<i>Tenor</i>	One week	One day	One day
<i>Interest rate</i>	By reference to prevailing market interest rates		TMA overnight CNH HIBOR fixing plus 50 bps
<i>Eligible collateral</i>	- Exchange Fund Bills and Notes (EFBN) - HKSAR Government bonds (HKGB) - RMB denominated bonds issued in Hong Kong by the Ministry of Finance of the People's Republic of China (CMOF) - RMB denominated bonds issued in Hong Kong by policy banks of the People's Republic of China, including Agricultural Development Bank of China, China Development Bank, and Export and Import Bank of China		
<i>Haircut on eligible collateral</i>	- EFBN and HKGB: 2% per year of remaining maturity, plus 2% (for cross-currency haircut) - CMOF and China policy bank bonds: 2% per year of remaining maturity, minimum 2%		
<i>Banks eligible ^A</i>	Als participating in RMB business (Participating Als)		
<i>Cut-off time for request ^B</i>	12:00 noon on each Hong Kong business day (not available on mainland China holidays)		3:00 pm on each Hong Kong business day
<i>Contact details</i>	Participating Als interested in borrowing RMB funds should contact the dealing room of the HKMA at 2878 8104 or Reuters dealing code EFHK		
<i>Settlement</i>	On T+1, RMB funds will be credited to the Participating Al's RMB RTGS account held with the Clearing Bank subject to the receipt of the securities by the HKMA before 4:00 pm		On T+0, RMB funds will be credited to the Participating Al's RMB RTGS account held with the Clearing Bank subject to the receipt of the securities by the HKMA before 4:00 pm
<i>Holiday</i>	If the value date or maturity date of the transaction falls on a Hong Kong or China holiday, it will be delayed to the next business day which is not a holiday in Hong Kong and China.		If the maturity date of the lending falls on a Hong Kong holiday, it will be delayed to the next Hong Kong business day.

^A Participating Als were required to have signed with the HKMA the Master Sale and Repurchase Agreement for the provision of liquidity assistance, including lender-of-last-resort support and RMB liquidity facility.

^B Requests for overnight funds received after the cut-off time of 3:00 pm would be considered on a case-by-case basis.

Source: Recreated from HKMA 2014a.

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